Lab Title:- RECONSTRUCTING SUSPECTED DRUG TRAFFICKING FROM CHROME WEB HISTORY

Student Name:- BADAMASI MUNTAKA MUHAMMAD

Student ID:- 2025FWSD11563

Course Name:- COMPUTER FORENSICS CASE STUDIES

Instructor Name:- COMMANDANT AMINU IDRIS

Date Of Submission:- 12/10/2026

Case scenario

Investigators received a Google Chrome History file from a device associated with a suspect. The case hypothesis states that the browser may have been used to advertise controlled drugs, communicate with an interested buyer, exchange payment information, verify a cryptocurrency transaction, and retain proof of payment. Your task is to assess the available history evidence and construct a defensible timeline. The case requires forensic interpretation of a supplied database, not interaction with live websites, accounts, messages, or wallets.

Lab Objectives

The objectives of the lab is to understand how to reconstruct digital evidence that might exist in any browser history using a tool like sqlitebrowser.

Tools and Resources Used

- Kali linux:- A penetration testing platform used for security assessments.

Kali linux tools

- file:- determine file type
- sqlitebrowser (SQLite Database Browser):- is a visual tool used to create, design and edit database files compatible with SQLite
- printf:- used for formatting and printing of data

Methology

- I power on the kali linux machine
- I download the **History** file using the wget command

- I made a copy of the file(History) and then obtain the hash value(MD5/SHA256) of the file(History)
- I upload the file(History) into the sqlitebrowser and then start my analysis.

Screenshot and Evidence

SCREENSHOT SHOWING SOME BASIC INFORMATION ABOUT THE HISTORY FILE



SCREENSHOT 1 (EVIDENCE 1)

SQL1* ×

```
1  select id, "url_ref_ID: " || url, datetime(visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime'), "From_visit:
2
```

	id	"url_ref_ID: " || url	0+strftime('%s','1601-01-…	om_visit: " || from_v	'Tran:' || transition	"Duration: " || (visit_duration/1000)
1	1	url_ref_ID: 1	2022-04-19 09:54:18	From_visit: 0	Tran:805306373	Duration: 0
2	2	url_ref_ID: 1	2022-04-19 09:54:19	From_visit: 0	Tran:805306368	Duration: 727

SQL1* ×

```
1  select  "url_id: " || id, url, title, datetime(last_visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from
2
```

	"url_id: " || id	url	title	me/1000000+strftime('%s','1601-01-01'), 'ur
1	url_id: 1	https://www.google.com/search?…	craigslist - Google Search	2022-04-19 09:54:19

SCREENSHOT 2 (EVIDENCE 2)

3	3	url_ref_ID: 2	2022-04-19 09:54:19	From_visit: 2	Tran:268435456	Duration: 0
4	4	url_ref_ID: 3	2022-04-19 09:54:19	From_visit: 3	Tran:-2147483648	Duration: 0
5	5	url_ref_ID: 4	2022-04-19 09:54:19	From_visit: 4	Tran:-1610612736	Duration: 2465
6	6	url_ref_ID: 5	2022-04-19 09:54:22	From_visit: 5	Tran:268435456	Duration: 0
7	7	url_ref_ID: 6	2022-04-19 09:54:22	From_visit: 6	Tran:-1610612736	Duration: 17456

SQL1* ×

```
1  select  "url_id: " || id, url, title, datetime(last_visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from urls where id=2 or id=3 or id=
2
```

	"url_id: " || id	url	title	me/1000000+strftime
1	url_id: 2	https://www.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events	2022-04-19 09:55:03
2	url_id: 3	https://geo.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events	2022-04-19 09:54:19
3	url_id: 4	https://baltimore.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events	2022-04-19 09:55:03
4	url_id: 5	https://accounts.craigslist.org/…	craigslist account	2022-04-19 09:54:39
5	url_id: 6	https://accounts.craigslist.org/…	craigslist - account log in	2022-04-19 09:54:22

SCREENSHOT 3 (EVIDENCE 3)

SQL 1* ×

```
1  select id, "url_ref_ID: " || url, datetime(visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime'), "From_visit:
2
```

	"url_ref_ID: " || url	:/1000000+strftime('%s','1601-01-01'), 'unix	"From_visit: " || from_visit	'Tran:' || transition	"Duration: " || (visit_duration/100(
8	url_ref_ID: 5	2022-04-19 09:54:39	From_visit: 7	Tran:805306375	Duration: 23439
9	url_ref_ID: 2	2022-04-19 09:55:03	From_visit: 8	Tran:268435456	Duration: 0
10	url_ref_ID: 4	2022-04-19 09:55:03	From_visit: 9	Tran:-1610612736	Duration: 2664

SQL 1* ×

```
1  "url_id: " || id, url, title, datetime(last_visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from urls where id=5  or id=2 or id=4;
```

	"url_id: " || id	url	title	me/1000000+strftime(
1	url_id: 2	https://www.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events	2022-04-19 09:55:03
2	url_id: 4	https://baltimore.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events	2022-04-19 09:55:03
3	url_id: 5	https://accounts.craigslist.org/login/home	craigslist account	2022-04-19 09:54:39

SCREENSHOT 4(EVIDENCE 4)

SQL 1* ×

```
1  select id, "url_ref_ID: " || url, datetime(visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime'), "Fro
2
```

	id	"url_ref_ID: " || url	0+strftime('%s','1601-01-|	om_visit: " || from_v	'Tran:' || transition	"Duration: " || (visit_duration/1000)
11	11	url_ref_ID: 7	2022-04-19 09:55:05	From_visit: 10	Tran:268435456	Duration: 0
12	12	url_ref_ID: 8	2022-04-19 09:55:05	From_visit: 11	Tran:-2147483648	Duration: 0
13	13	url_ref_ID: 9	2022-04-19 09:55:05	From_visit: 12	Tran:-1610612736	Duration: 4273
14	14	url_ref_ID: 10	2022-04-19 09:55:10	From_visit: 13	Tran:805306375	Duration: 5599
15	15	url_ref_ID: 11	2022-04-19 09:55:15	From_visit: 14	Tran:805306375	Duration: 177674

SQL 1* ×

```
1  tle, datetime(last_visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from urls where id=7 or id=8 or id=9 or id=10 or id=11;
2
```

	"url_id: " || id	url	title	me/1000000+strftime('
1	url_id: 7	https://post.craigslist.org/c/bal	baltimore | choose type	2022-04-19 09:55:05
2	url_id: 8	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5	baltimore | posting confirmation	2022-04-19 09:58:22
3	url_id: 9	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=type	baltimore | choose type	2022-04-19 09:55:05
4	url_id: 10	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=cat	baltimore | choose category	2022-04-19 09:55:10
5	url_id: 11	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=edit	baltimore | create posting	2022-04-19 09:55:15

SCREENSHOT 5 (EVIDENCE 5)

16	16	url_ref_ID: 12	2022-04-19 09:58:13	From_visit: 15	Tran:805306375	Duration: 4389
17	17	url_ref_ID: 13	2022-04-19 09:58:17	From_visit: 16	Tran:805306375	Duration: 1649
18	18	url_ref_ID: 14	2022-04-19 09:58:19	From_visit: 17	Tran:805306375	Duration: 3199

SQL 1* ×

```
1  select  "url_id: " || id, url, title, datetime(last_visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from urls where id
2
```

	"url_id: " || id	url	title	me/1000000+strftime(
1	url_id: 12	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=geoverify	baltimore | add map	2022-04-19 09:58:13
2	url_id: 13	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=editimage	baltimore | create posting	2022-04-19 09:58:17
3	url_id: 14	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=preview	baltimore | create posting	2022-04-19 09:58:19

SCREENSHOT 6 (EVIDENCE 6)

19	19	url_ref_ID: 8	2022-04-19 09:58:22	From_visit: 18	Tran:805306375	Duration: 4989
20	20	url_ref_ID: 15	2022-04-19 09:58:27	From_visit: 19	Tran:805306368	Duration: 5597

SQL1* ×

```
1  select  "url_id: " || id, url, title, datetime(last_visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from u
2
```

	"url_id: " || id	url	title	me/1000000+strftime('
1	url_id: 8	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5	baltimore | posting confirmation	2022-04-19 09:58:22
2	url_id: 15	https://post.craigslist.org/manage/7473121658	baltimore | manage posting	2022-04-19 09:58:27

SCREENSHOT 7 (EVIDENCE 7)

SQL1* ×

```
1  select id, "url_ref_ID: " || url, datetime(visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime'), "Fro
2
```

	id	"url_ref_ID: " || url	0+strftime('%s','1601-01-	om_visit: " || from_v	'Tran:' || transition	"Duration: " || (visit_duration/1000)
21	21	url_ref_ID: 16	2022-04-19 09:58:31	From_visit: 0	Tran:805306368	Duration: 0
22	22	url_ref_ID: 17	2022-04-19 09:58:38	From_visit: 0	Tran:838860805	Duration: 0
23	23	url_ref_ID: 17	2022-04-19 09:58:39	From_visit: 0	Tran:805306368	Duration: 842
24	24	url_ref_ID: 18	2022-04-19 09:58:40	From_visit: 23	Tran:268435456	Duration: 0
25	25	url_ref_ID: 19	2022-04-19 09:58:40	From_visit: 24	Tran:-2147483648	Duration: 0
26	26	url_ref_ID: 20	2022-04-19 09:58:40	From_visit: 25	Tran:-2147483648	Duration: 0

SQL1* ×

```
1  ), 'unixepoch', 'localtime') from urls where id=16 or id=17 or id=18 or id=19 or id=20;
2
```

	"url_id: " || id	url	title	0+strftime('%s','1601-
1	url_id: 16	https://baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/…	cheaper than Rx supplements - health and beauty …	2022-04-19 09:58:31
2	url_id: 17	https://www.google.com/search?q=gmail&rlz=1C1ONGR_enUS989US989&oq=gmail&aqs=chrome.…	gmail - Google Search	2022-04-19 09:58:39
3	url_id: 18	https://www.google.com/gmail/	Gmail: Free, Private & Secure Email | Google …	2022-04-19 09:58:40
4	url_id: 19	https://mail.google.com/mail/	Inbox (12) - unsub.fscs@gmail.com - Gmail	2022-04-19 09:59:29
5	url_id: 20	https://mail.google.com/mail/u/0/	Inbox (12) - unsub.fscs@gmail.com - Gmail	2022-04-19 09:59:29

SCREENSHOT 8 (EVIDENCE 8)

27	27	url_ref_ID: 21	2022-04-19 09:58:40	From_visit: 26	Tran:-2147483648	Duration: 0
28	28	url_ref_ID: 22	2022-04-19 09:58:40	From_visit: 27	Tran:-2147483648	Duration: 0
29	29	url_ref_ID: 23	2022-04-19 09:58:40	From_visit: 28	Tran:-2147483648	Duration: 0
30	30	url_ref_ID: 24	2022-04-19 09:58:40	From_visit: 29	Tran:-1610612736	Duration: 0

SQL 1* x

```
1  "url_id: " || id, url, title, datetime(last_visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from urls where id=21 or id=22 or
2
```

	"url_id:" || id	url	title	me/1000000+strftime('
1	url_id: 21	https://accounts.google.com/ServiceLogin?…	Gmail: Free, Private & Secure Email | Google Workspace	2022-04-19 09:58:40
2	url_id: 22	https://mail.google.com/intl/en/mail/help/about.html	Gmail: Free, Private & Secure Email | Google Workspace	2022-04-19 09:58:40
3	url_id: 23	https://www.google.com/intl/en/mail/help/about.html	Gmail: Free, Private & Secure Email | Google Workspace	2022-04-19 09:58:40
4	url_id: 24	https://www.google.com/gmail/about/	Gmail: Free, Private & Secure Email | Google Workspace	2022-04-19 09:58:40

SCREENSHOT 9 (EVIDENCE 9)

SQL 1* ×

```
1  select id, "url_ref_ID: " || url, datetime(visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime'), "Fro
2
```

	id	"url_ref_ID: " || url	0+strftime('%s','1601-01-	om_visit: " || from_v	'Tran:' || transition	"Duration: " || (visit_duration/1000)
31	31	url_ref_ID: 25	2022-04-19 09:58:43	From_visit: 0	Tran:268435456	Duration: 0
32	32	url_ref_ID: 26	2022-04-19 09:58:43	From_visit: 31	Tran:-2147483648	Duration: 0
33	33	url_ref_ID: 27	2022-04-19 09:58:43	From_visit: 32	Tran:-1610612736	Duration: 0
34	34	url_ref_ID: 27	2022-04-19 09:58:44	From_visit: 0	Tran:805306368	Duration: 0
35	35	url_ref_ID: 28	2022-04-19 09:58:44	From_visit: 0	Tran:805306368	Duration: 0

SQL 1* ×

```
1   "url_id: " || id, url, title, datetime(last_visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime')
2
```

	"url_id: " || id	url	title	me/1000000+strftime('
1	url_id: 25	https://accounts.google.com/AccountChooser/signinchooser?…	Gmail	2022-04-19 09:58:43
2	url_id: 26	https://accounts.google.com/AccountChooser?…	Gmail	2022-04-19 09:58:43
3	url_id: 27	https://accounts.google.com/ServiceLogin?…	Gmail	2022-04-19 09:58:44
4	url_id: 28	https://accounts.google.com/signin/v2/identifier?…	Gmail	2022-04-19 09:58:44

SCREENSHOT 10 (EVIDENCE 10)

36	36	url_ref_ID: 29	2022-04-19 09:59:26	From_visit: 0	Tran:805306368	Duration: 2967
37	37	url_ref_ID: 30	2022-04-19 09:59:29	From_visit: 36	Tran:268435456	Duration: 0
38	38	url_ref_ID: 31	2022-04-19 09:59:29	From_visit: 37	Tran:-2147483648	Duration: 0
39	39	url_ref_ID: 32	2022-04-19 09:59:29	From_visit: 38	Tran:-2147483648	Duration: 0

SQL 1* ×

```
1  "url_id: " || id, url, title, datetime(last_visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from urls where id=
2
```

	"url_id: " || id	url	title	me/1000000+strftime("
1	url_id: 29	https://accounts.google.com/signin/v2/challenge/pwd?…	Gmail	2022-04-19 09:59:26
2	url_id: 30	https://accounts.google.com/CheckCookie?…	Inbox (12) - unsub.fscs@gmail.com - Gmail	2022-04-19 09:59:29
3	url_id: 31	https://mail.google.com/accounts/SetOSID?…	Inbox (12) - unsub.fscs@gmail.com - Gmail	2022-04-19 09:59:29
4	url_id: 32	https://accounts.youtube.com/accounts/SetSID?…	Inbox (12) - unsub.fscs@gmail.com - Gmail	2022-04-19 09:59:29

SCREENSHOT 11 (EVIDENCE 11)

	id	"url_ref_ID: " || url	e/1000000+strftime('%s','1601-01-01'), 'unix	"From_visit: " || from_visit	'Tran:' || transition	"Duration: " || (visit_duration/1000)
40	40	url_ref_ID: 19	2022-04-19 09:59:29	From_visit: 39	Tran:-2147483648	Duration: 0
41	41	url_ref_ID: 20	2022-04-19 09:59:29	From_visit: 40	Tran:-2147483648	Duration: 0
42	42	url_ref_ID: 33	2022-04-19 09:59:31	From_visit: 41	Tran:1610612736	Duration: 0
43	43	url_ref_ID: 34	2022-04-19 10:04:12	From_visit: 0	Tran:268435456	Duration: 0
44	44	url_ref_ID: 35	2022-04-19 10:04:35	From_visit: 43	Tran:1073741824	Duration: 0
45	45	url_ref_ID: 36	2022-04-19 10:04:46	From_visit: 44	Tran:1073741824	Duration: 0

SQL 1* ×

```
1  id, url, title, datetime(last_visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from urls where id=19 or id=20 or id=33 or id=34 o
2
```

	"url_id: " || id	url	title	00+strftime('%s','1601-
1	url_id: 19	https://mail.google.com/mail/	Inbox (12) - unsub.fscs@gmail.com - Gmail	2022-04-19 09:59:29
2	url_id: 20	https://mail.google.com/mail/u/0/	Inbox (12) - unsub.fscs@gmail.com - Gmail	2022-04-19 09:59:29
3	url_id: 33	https://mail.google.com/mail/u/0/#inbox	Inbox (12) - unsub.fscs@gmail.com - Gmail	2022-04-19 10:05:59
4	url_id: 34	https://mail.google.com/mail/u/0/#inbox/…	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail	2022-04-19 10:04:52
5	url_id: 35	https://mail.google.com/mail/u/0/#inbox/…	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail	2022-04-19 10:04:35
6	url_id: 36	https://mail.google.com/mail/u/0/#inbox/…	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail	2022-04-19 10:04:46

SCREENSHOT 12 (EVIDENCE 12)

	id	"url_ref_ID: " || url	e/1000000+strftime('%s','1601-01-01'), 'unix	"From_visit: " || from_visit	'Tran:' || transition	"Duration: " || (visit_duration/1000)
46	46	url_ref_ID: 34	2022-04-19 10:04:52	From_visit: 45	Tran:1610612736	Duration: 0
47	47	url_ref_ID: 33	2022-04-19 10:05:59	From_visit: 0	Tran:805306368	Duration: 0
48	48	url_ref_ID: 37	2022-04-19 10:56:47	From_visit: 0	Tran:268435456	Duration: 0
49	49	url_ref_ID: 38	2022-04-19 10:56:52	From_visit: 0	Tran:268435456	Duration: 0
50	50	url_ref_ID: 39	2022-04-19 10:56:52	From_visit: 49	Tran:-1610612736	Duration: 18926

SQL 1* ×

```
1  id, url, title, datetime(last_visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from urls where id=33 or id=34 or id=37 or id=38 or id=3
2
```

	"url_id: " || id	url	title	me/1000000+strftime("
1	url_id: 33	https://mail.google.com/mail/u/0/#inbox	Inbox (12) - unsub.fscs@gmail.com - Gmail	2022-04-19 10:05:59
2	url_id: 34	https://mail.google.com/mail/u/0/#inbox/…	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail	2022-04-19 10:04:52
3	url_id: 37	https://mail.google.com/mail/u/0/#inbox/…	cheaper than Rx supplements -- txid: …	2022-04-19 11:02:56
4	url_id: 38	https://www.google.com/url?q=https://imgur.com/…	Imgur: The magic of the Internet	2022-04-19 10:56:52
5	url_id: 39	https://imgur.com/dTgrkP7	Imgur: The magic of the Internet	2022-04-19 10:56:52

SCREENSHOT 13 (EVIDENCE 13)

SQL 1* ×

```
1  select id, "url_ref_ID: " || url, datetime(visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime'), "Fro
2
```

	id	"url_ref_ID: " || url	0+strftime('%s','1601-01-…	om_visit: " || from_v	'Tran:' || transition	"Duration: " || (visit_duration/1000)
51	51	url_ref_ID: 40	2022-04-19 10:58:59	From_visit: 0	Tran:838860805	Duration: 0
52	52	url_ref_ID: 40	2022-04-19 10:59:00	From_visit: 0	Tran:805306368	Duration: 1026
53	53	url_ref_ID: 41	2022-04-19 10:59:01	From_visit: 52	Tran:805306368	Duration: 0
54	54	url_ref_ID: 41	2022-04-19 10:59:02	From_visit: 0	Tran:805306368	Duration: 18679
55	55	url_ref_ID: 42	2022-04-19 10:59:06	From_visit: 54	Tran:268435456	Duration: 0
56	56	url_ref_ID: 43	2022-04-19 10:59:06	From_visit: 55	Tran:-1610612736	Duration: 0
57	57	url_ref_ID: 43	2022-04-19 10:59:06	From_visit: 54	Tran:805306368	Duration: 0
58	58	url_ref_ID: 44	2022-04-19 10:59:21	From_visit: 54	Tran:805306368	Duration: 0
59	59	url_ref_ID: 45	2022-04-19 10:59:38	From_visit: 0	Tran:838860805	Duration: 0
60	60	url_ref_ID: 45	2022-04-19 10:59:39	From_visit: 0	Tran:805306368	Duration: 7703

SQL 1* ×

```
1  id, url, title, datetime(last_visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from urls where id=40 or id=41 or id=42 or id=43 or id=44 or id=45;
2
```

	"url_id: " || id	url	title	)00+strftime('%s','160
1	url_id: 40	https://www.google.com/search?…	blockchain explorer - Google Search	2022-04-19 10:59:00
2	url_id: 41	https://www.blockchain.com/explorer	Blockchain Explorer - Search the Blockchain | BTC | ETH | BCH	2022-04-19 10:59:02
3	url_id: 42	https://www.blockchain.com/search?…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 | Blockchain Explorer	2022-04-19 10:59:06
4	url_id: 43	https://www.blockchain.com/btc/tx/…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 | Blockchain Explorer	2022-04-19 10:59:06
5	url_id: 44	https://www.blockchain.com/btc/…	Address: 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt | Blockchain Explorer	2022-04-19 10:59:21
6	url_id: 45	https://www.google.com/search?…	kraken - Google Search	2022-04-19 10:59:39

SCREENSHOT 14 (EVIDENCE 14)

	id	"url_ref_ID: " || url	0+strftime('%s','1601-01-(	om_visit: " || from_v	'Tran:' || transition	"Duration: " || (visit_duration/1000)
61	61	url_ref_ID: 46	2022-04-19 10:59:47	From_visit: 60	Tran:805306368	Duration: 0
62	62	url_ref_ID: 46	2022-04-19 10:59:47	From_visit: 0	Tran:805306368	Duration: 0
63	63	url_ref_ID: 47	2022-04-19 11:00:21	From_visit: 0	Tran:805306368	Duration: 104425
64	64	url_ref_ID: 48	2022-04-19 11:01:11	From_visit: 63	Tran:805306368	Duration: 0
65	65	url_ref_ID: 49	2022-04-19 11:01:12	From_visit: 63	Tran:805306368	Duration: 0
66	66	url_ref_ID: 50	2022-04-19 11:01:21	From_visit: 63	Tran:805306368	Duration: 0

SQL1* ×

```
1  id, url, title, datetime(last_visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from urls where id=46 or id=47 or id=48 or id=49 or id=50;
2
```

	"url_id: " || id	url	title	00+strftime('%s','1601
1	url_id: 46	https://www.kraken.com/en-us/sign-in	Sign in to Kraken - Kraken | Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)	2022-04-19 10:59:47
2	url_id: 47	https://www.kraken.com/en-us/device-approval	Approve new device - Kraken | Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)	2022-04-19 11:00:21
3	url_id: 48	https://www.kraken.com/u/trade?i=true		2022-04-19 11:01:11
4	url_id: 49	https://www.kraken.com/u/instant	Kraken | Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)	2022-04-19 11:01:12
5	url_id: 50	https://www.kraken.com/u/funding	41,543.80 USD - Kraken - Funding - Overview	2022-04-19 11:01:21

SCREENSHOT 15 (EVIDENCE 15)

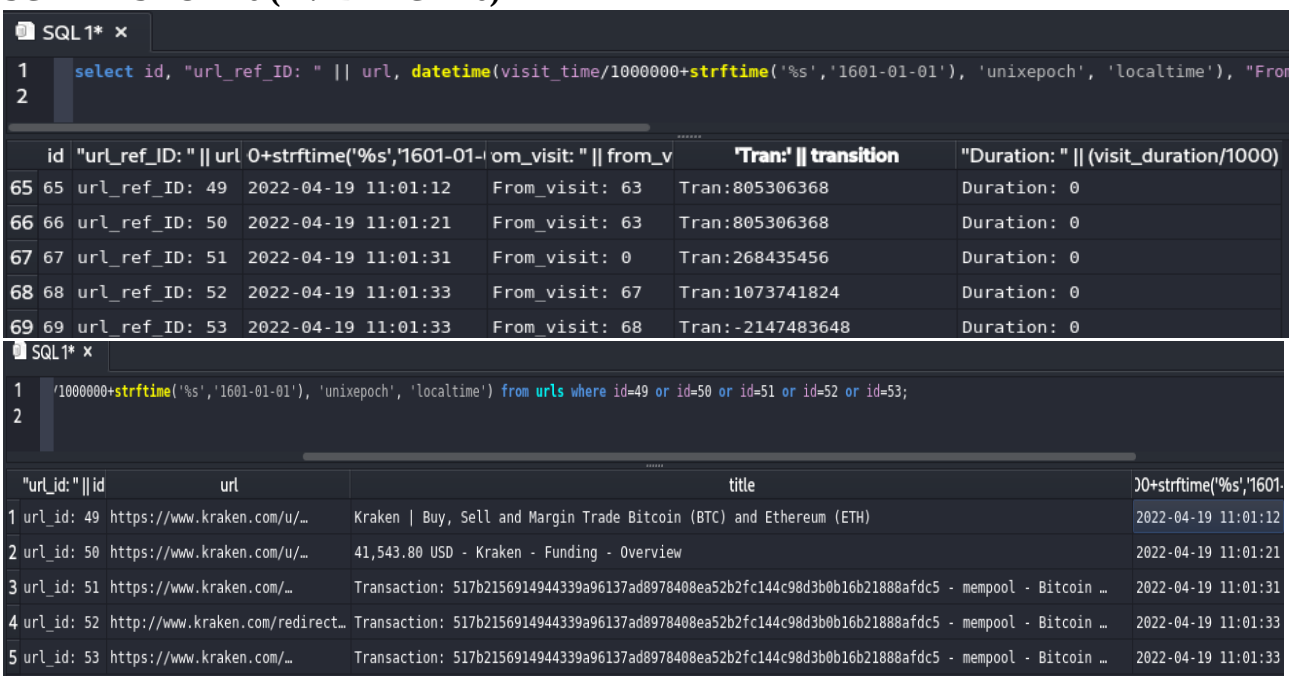
67	67	url_ref_ID: 51	2022-04-19 11:01:31	From_visit: 0	Tran:268435456	Duration: 0
68	68	url_ref_ID: 52	2022-04-19 11:01:33	From_visit: 67	Tran:1073741824	Duration: 0
69	69	url_ref_ID: 53	2022-04-19 11:01:33	From_visit: 68	Tran:-2147483648	Duration: 0
70	70	url_ref_ID: 54	2022-04-19 11:01:33	From_visit: 69	Tran:-1610612736	Duration: 0

SQL1* ×

```
1  "url_id: " || id, url, title, datetime(last_visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from urls where id=51 or id=52 or id=53 or id=54;
2
```

	"url_id: " || id	url	title	00+strftime('%s','1601-
1	url_id: 51	https://www.kraken.com/…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer	2022-04-19 11:01:31
2	url_id: 52	http://www.kraken.com/…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer	2022-04-19 11:01:33
3	url_id: 53	https://www.kraken.com/…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer	2022-04-19 11:01:33
4	url_id: 54	https://mempool.space/tx/…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer	2022-04-19 11:01:33

SCREENSHOT 16 (EVIDENCE 16)



SQL 1* ×

```
1  select id, "url_ref_ID: " || url, datetime(visit_time/1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime'), "Fro
2
```

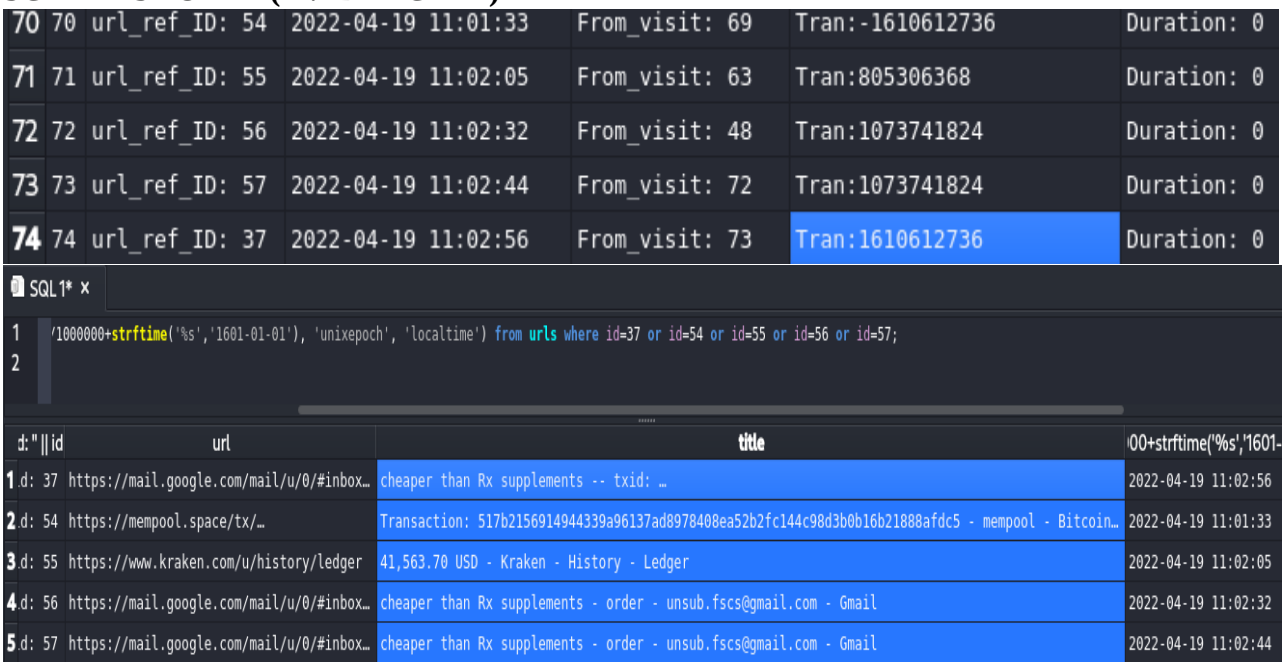
	id	"url_ref_ID: " || url	0+strftime('%s','1601-01-…	om_visit: " || from_v	'Tran:' || transition	"Duration: " || (visit_duration/1000)
65	65	url_ref_ID: 49	2022-04-19 11:01:12	From_visit: 63	Tran:805306368	Duration: 0
66	66	url_ref_ID: 50	2022-04-19 11:01:21	From_visit: 63	Tran:805306368	Duration: 0
67	67	url_ref_ID: 51	2022-04-19 11:01:31	From_visit: 0	Tran:268435456	Duration: 0
68	68	url_ref_ID: 52	2022-04-19 11:01:33	From_visit: 67	Tran:1073741824	Duration: 0
69	69	url_ref_ID: 53	2022-04-19 11:01:33	From_visit: 68	Tran:-2147483648	Duration: 0

SQL 1* ×

```
1  '1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from urls where id=49 or id=50 or id=51 or id=52 or id=53;
2
```

	"url_id: " || id	url	title	00+strftime('%s','1601-
1	url_id: 49	https://www.kraken.com/u/…	Kraken | Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)	2022-04-19 11:01:12
2	url_id: 50	https://www.kraken.com/u/…	41,543.80 USD - Kraken - Funding - Overview	2022-04-19 11:01:21
3	url_id: 51	https://www.kraken.com/…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin …	2022-04-19 11:01:31
4	url_id: 52	http://www.kraken.com/redirect…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin …	2022-04-19 11:01:33
5	url_id: 53	https://www.kraken.com/…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin …	2022-04-19 11:01:33

SCREENSHOT 17 (EVIDENCE 17)



70	70	url_ref_ID: 54	2022-04-19 11:01:33	From_visit: 69	Tran:-1610612736	Duration: 0
71	71	url_ref_ID: 55	2022-04-19 11:02:05	From_visit: 63	Tran:805306368	Duration: 0
72	72	url_ref_ID: 56	2022-04-19 11:02:32	From_visit: 48	Tran:1073741824	Duration: 0
73	73	url_ref_ID: 57	2022-04-19 11:02:44	From_visit: 72	Tran:1073741824	Duration: 0
74	74	url_ref_ID: 37	2022-04-19 11:02:56	From_visit: 73	Tran:1610612736	Duration: 0

SQL 1* ×

```
1  '1000000+strftime('%s','1601-01-01'), 'unixepoch', 'localtime') from urls where id=37 or id=54 or id=55 or id=56 or id=57;
2
```

	d: " || id	url	title	00+strftime('%s','1601-
1	d: 37	https://mail.google.com/mail/u/0/#inbox…	cheaper than Rx supplements -- txid: …	2022-04-19 11:02:56
2	d: 54	https://mempool.space/tx/…	Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin…	2022-04-19 11:01:33
3	d: 55	https://www.kraken.com/u/history/ledger	41,563.70 USD - Kraken - History - Ledger	2022-04-19 11:02:05
4	d: 56	https://mail.google.com/mail/u/0/#inbox…	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail	2022-04-19 11:02:32
5	d: 57	https://mail.google.com/mail/u/0/#inbox…	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail	2022-04-19 11:02:44

Analysis and Discussion

The table below summarize my findings based on the screenshot/evidences I obtain which I have provide in the screenshots/evidence section provided above

VISIT(URL ID)	FINDINGS	TIMESTAMP	SCREENSHOT NO.
1-2 (1)	The suspect search for a website(craigslist) in the google chrome browser	2022-04-19 09:54:18(V1) & 2022-04-19 09:54:19(V2)	1
3-7(2)	The suspect click on the result of the search he made	2022-04-19 09:54:19	2
3-7(3)	After the suspect suspect click on the result he was redirected	2022-04-19 09:54:19	2
3-7(4)	The suspect was redirected to Baltimore whisch is just like a subset of (craigslist)	2022-04-19 09:54:19	2
3-7(5)	A login page pop-up for the suspect to login	2022-04-19 09:54:22	2
3-7(6)	The suspect input his login details that landed him successfully into his account on the crag	2022-04-19 09:54:22	2
11-15(7)	The suspect start to create a post after he has been login into his account and was asked to choose a type	2022-04-19 09:55:05	4
11-15(8)	The suspect submitted a type for his post that result in a posting confirmation	2022-04-19 09:55:05(V12) & 2022-04-19 09:58:22(V19)	4
11-15(9)	Type selection for the post was successfully submitted	2022-04-19 09:55:05	4
11-15(10)	The suspect continues with the post creation whereby he was prompt to choose a category	2022-04-19 09:55:10	4
11-15(11)	The suspect was creating his posting content	2022-04-19 09:55:15	4
16-18(12)	The suspect added a map to his post	2022-04-19 09:58:13	5

16-18(13)	The suspect was editing an image on his post	2022-04-19 09:58:17	5
16-18(14)	The suspect was reviewing his post be he posted it	2022-04-19 09:58:19	5
19-20(15)	The suspect was managing (editing,updating/deleting) his post	2022-04-19 09:58:27	6
21-16(16)	The suspect view his post after he has published it	2022-04-19 09:58:31	7
21-16 (17)	The suspect make a search(gmail) on google chrome browser	2022-04-19 09:58:38(V22) & 2022-04-19 09:58:39(V23)	7
21-16 (18)	The suspect click on the gmail search result	2022-04-19 09:58:40	7
21-16 (19)	A email account(unsub.fscs@gmail.com) inbox was visible to the suspect	2022-04-19 09:58:40(V25) & 2022-04-19 09:59:29(V40)	7
21-16 (20)	The suspect was still inside the email account	2022-04-19 09:58:40(V26) & 2022-04-19 09:59:29(V41)	7
27-30(21)	The suspect was still inside the Gmail website but was redirected to ServiceLogin	2022-04-19 09:58:40	8
27-30 (22&23)	The suspect was still inside the Gmail website but was again redirected to a page about.html	2022-04-19 09:58:40	8
27-30 (24)	The suspect is still inside the Gmail website but on the about page	2022-04-19 09:58:40	8
31-35(25)	The Gmail sign in process started automatically and a prompt was display to the suspect to choose an account	2022-04-19 09:58:43	9
31-35 (26)	The Gmail sign process continues	2022-04-19 09:58:43	9
31-35 (27)	The suspect was redirected to ServiceLogin to continue in order to input an email address	2022-04-19 09:58:43 (V33) &	9

		2022-04-19 09:58:44 (V34)	
31-35 (28)	The email address entered by the suspect undergoes a verification	2022-04-19 09:58:44	9
36-39(29)	The suspect was prompt for a password after the verification of the email address	2022-04-19 09:59:26	10
36-39 (30)	The password entered by the suspect was redirected to CheckCookie?continue=https%3A which in return shows that the suspect was successfully login into the email account()	2022-04-19 09:59:29	10
36-39 (31)	Recreating the gmail accountSETOSID?authuser=0 after successful login to the email account	2022-04-19 09:59:29	10
36-39 (32)	Recreating account session (SetSID/ssdc)	2022-04-19 09:59:29	10
40-45(33)	The email account inbox was visible to the suspect	2022-04-19 09:59:31 (V42) & 2022-04-19 10:05:59 (V47)	11
40-45 (34)	The suspect opens an order message in the email account inbox	2022-04-19 10:04:12 (V43) & 2022-04-19 10:04:52 (V46)	11
40-45 (35&36)	The suspect was still inside the order message he opens	2022-04-19 10:04:35 & 2022-04-19 10:04:46	11
46-50(37)	The suspect opens another message in the inbox in which inside that message there was a trasactionction id(txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5)	2022-04-19 10:56:47 (V48) & 2022-04-19 11:02:56 (V74)	12
46-50 (38)	The suspect click a link in the message he opens which took him to a website imgur: the magic ot the internet	2022-04-19 10:56:52	12

46-50 (39)	The suspect was still at the website imgur:the magic of the internet	2022-04-19 10:56:52	12
51-60(40)	The suspect made a google search on blockchain explorer using the google chrome browser	2022-04-19 10:58:59 (V51) & 2022-04-19 10:59:00 (V52)	13
51-60 (41)	The suspect click on the google search result of the blockchain explorer which landed the suspect into the blockchain explorer website.	2022-04-19 10:59:01 (V53) & 2022-04-19 10:59:02 (V54)	13
51-60 (42)	The suspect search for the transaction id(517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5) he received inside the inbox of the email account(unsub.fscs@gmail.com) earlier in the blockchain explorer website	2022-04-19 10:59:06	13
51-60 (43)	The search returns the details about the transaction id(517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5) the suspect made a search on	2022-04-19 10:59:06 (V56) & 2022-04-19 10:59:06 (V57)	13
51-60 (44)	The suspect navigate to the details and the a btc address id (517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5) was visible which is the same as the one in the inbox of the email address he opens earlier	2022-04-19 10:59:21	13
51-60 (45)	The suspect made a google search on kraken using the google chrome browser	2022-04-19 10:59:38 (V59) & 2022-04-19 10:59:39 (V60)	13
61-66(46)	The suspect click on the search result for kraken and then made an attempt to sign into the kraken website	2022-04-19 10:59:47 (V61) & 2022-04-19 10:59:47 (V62)	14

61-66 (47)	A prompt appear to the suspect asking the suspect for a device approval which I think is because the suspect used a different device for signing in	2022-04-19 11:00:21	14
61-66 (48)	The suspect was successfully login into kraken website	2022-04-19 11:01:11	14
61-66 (49)	The suspect navigate to (instance) which is just like an overview of the kraken website after successful login	2022-04-19 11:01:12	14
61-66 (50)	The suspect navigate to funding in the kraken website which display an amount of 41,543.80 USD	2022-04-19 11:01:21	14
67-70(51)	Atb the kraken website the transaction id(517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5) that the suspect made a search on in the blockchain explorer was visible and also a redirection was made to the mempool bitcoin explorer	2022-04-19 11:01:31	15
67-70 (52 & 53)	Redirection to mempool bitcoin explorer in progress	2022-04-19 11:01:33 & 2022-04-19 11:01:33	15
67-70 (54)	The suspect was successfully landed at the mempool bitcoin explorer which shows the same transaction id(517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5)	2022-04-19 11:01:33	15
71-74(55)	The suspect went back to “krakn.com/u/history/ledger” where a different amount (41,563.70 USD) was visible which is different from the one display earlier (41,543.80 USD)	2022-04-19 11:02:05	17
71-74 (56)	The suspect went back to the order message in the email account he opens earlier	2022-04-19 11:02:32	17
71-74 (57)	The suspect remains inside the order message	2022-04-19 11:02:44	17

Conclusion

In conclusion this lab is an interesting lab that has laid a very strong foundation on how to reconstruct digital evidences stored in a browser history.

Reference

➢ICDFA course Material (SBT-DF204(computer forensic case studies) by COMMANDANT AMINU IDRIS
